COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	**SECTION:**	CPE21S2
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ACTIVITY NO. 7.1 Using Sorting Algorithms

1. PROCEDURE OUTPUT

What procedures did you perform?

1. What is a sorting algorithm?

A sorting algorithm is a method used to arrange data in a specific order. It helps make data easier to read and manage. Sorting can be done from smallest to largest or largest to smallest. It can also arrange words alphabetically from A to Z.

2. Where can sorting algorithms be used?

The fundamental tools that were used to rearrange a collection of items into a specific order, such as numerical or alphabetical, to make the data significantly easier to search, optimize, and analyze.

3. Explain the different types of sorting algorithms.

Algorithms	Definitions	Difference
Bubble Sort	Is a comparison-based sorting algorithm that reads through the list repeatedly, compares each pair of adjacent elements, and swaps them if they are in the wrong order.	Worst-Case: $O(n^2)$ Best -Case: O(n) Method: Adjacent Swapping
Selection Sort	A sorting algorithm that divides the list into a sorted and an unsorted portion and repeatedly selects the minimum (or maximum) element from the unsorted portion to place at the end of the sorted portion.	Worst-Case: $O(n^2)$ Best-Case: $O(n^2)$ Method: Repeated min/max selection

Insertion Sort	A sorting algorithm that takes its time by building the sorted list one at a time by taking each element from the unsorted portion and placing it in its correct position within the already sorted part.	Worst-Case: $O(n^2)$ Best-Case: O(n) Method: Sequential insertion
Merge Sort	Is a sorting algorithm that splits the list into halves until each sublist contains a single element, then merges the sublists back together in sorted order.	Worst-Case: O(n log n) Best-Case: O(n log n) Method: Divide and Conquer (split/merge)
Quick Sort	An algorithm that selects a pivot element, divides the remaining elements into two sublists (those less than the pivot and those greater), and repeatedly sorts each sublist.	Worst-Case: $O(n^2)$ Best-Case: O(n log n) Method: Division
Heap Sort	A sorting algorithm that first builds a binary max-heap (or min-heap) from the input data, then repeatedly extracts the maximum (or minimum) element from the heap and rebuilds the heap until the list is sorted.	Worst-Case: O(n log n) Best-Case: O(n log n) Method: Heap Extraction

4. Give sample programs in C++ that use sorting algorithms, specifically selection sort, insertion sort, and bubble sort. Explain how the program work.

Selection Sort
Example: A teacher ranking the top 3 scorers out of 30 exam papers for an award ceremony **Info about the program: Repeatedly scans the remaining papers to find the highest score, then places it into the next position, one at a time** **Explanation: It's selection sort because each round *selects* the best (highest) remaining value and locks it in with a single swap — ideal when you only need the top few results and can stop early.**
Insertion Sort
Example: A librarian shelving returned books one at a time into an already-alphabetized shelf **Info about the program: Takes each new book and compares it to the sorted ones already on the shelf, sliding it into the correct position** **Explanation: It's insertion sort because it *inserts* each new item into its proper place among already-sorted data — efficient when the data is already mostly sorted and items arrive one at a time.**
Bubble Sort
Example: A PE teacher lining up students by height, having neighboring students compare and swap **Info about the program: Repeatedly compares adjacent pairs and swaps them if they're out of order, sweeping through until everyone is in place** **Explanation: It's bubble sort because it works through repeated *adjacent* comparisons and swaps, causing the tallest (or highest) values to gradually "bubble up" toward their correct position.**

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

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3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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